# Watermarking Baseline Measurements on Code Generation

## 1 Research question and measured scope

The study measures the detection and functional correctness of existing watermarking methods on HumanEval, the MBPP test split, and the LiveCodeBench release-v1 code-generation test split. The methods are KGW, SemStamp, SWEET, CodeWM, and SeqMark. The tables preserve existing measurements. They do not establish a new watermark method or a cross-dataset winner.

The full splits contain 164, 500, and 400 tasks, respectively. Several CodeWM runs cover only subsets. SeqMark has complete HumanEval and MBPP evaluations for two encoders; its LiveCodeBench execution was cancelled and contributes no complete benchmark result. Partial and superseded executions remain available as historical evidence.

## 2 Methods and experimental conventions

KGW biases a context-seeded token green list. SWEET applies a green-list bias at positions exceeding an entropy threshold. SemStamp rejection-samples line segments using keyed regions of sentence embeddings. CodeWM learns a context-dependent token pairing and an embedding switch. SeqMark applies mean-centred semantic rejection sampling to the complete extracted program, which is treated as one sequence. The SeqMark detector reuses the saved generation centre for both watermarked and unwatermarked programs.

The shared model in the main comparison is Qwen3.5-4B. KGW, SemStamp, SWEET, and the CodeWM chat runs use temperature 0.7, nucleus probability 0.8, top-$k = 20$, a 2000-token generation budget, and random seed 42. Thinking is disabled. Dataset-level prompts request Python without comments or docstrings. KGW and SWEET use green fraction 0.5 and bias 8. SemStamp uses line segmentation. CodeWM is measured at biases 2 and 8 with switch threshold 0.7 in the Qwen comparison. Its original-paper reproduction is separate.

SeqMark uses temperature 1.2 for centre samples, candidate programs, and paired controls, unrestricted nucleus sampling, disabled top-$k$, repetition penalty 1.05, 50 centre samples, and at most 50 rejection trials. A search over 100 MBPP training tasks selected five hash dimensions and green fraction 0.1. The same configuration applies to Gemma and SBERT on both test splits. Those training tasks are disjoint from the MBPP test split; HumanEval does not select the parameters. Gemma denotes the embeddinggemma-300m encoder; SBERT denotes the SemStamp sentence encoder. Their native tokenization, pooling, and truncation remain part of the encoder comparison.

These sampling and implementation differences are experimental conditions, so the five-method table is not a controlled comparison varying only the watermark algorithm. Every run uses one recorded generation seed; the tables do not estimate variation over independent seeds.

## 3 Evaluation protocol

Detection receives the extracted Python program. The positive class contains watermarked programs; the negative class contains paired unwatermarked model generations. Extraction failures are unscoreable. A method can also return a null detector score after successful extraction, so scored and unscoreable counts need not sum to the task count.

For target false-positive rate $q$, the reported operating point is

$$\mathrm{TPR}_q = \max_{t:\mathrm{FPR}(t)\le q} \mathrm{TPR}(t), \qquad \text{positive if } z \ge t.$$

This definition selects the best attainable deterministic threshold under the false-positive constraint. Equal scores remain together. The calculation uses no interpolation or randomized tie splitting. Positive and negative rates use their respective scoreable-class denominators. AUROC denotes the area under the receiver operating characteristic curve.

Functional evaluation uses the benchmark's saved code extraction and retains the task cohort independently of detector scoreability. The chat-protocol pass@1 values use one generated program per task. LiveCodeBench evaluation uses the recorded cap of 20 tests per task, with inputs larger than 10 MB excluded during test preparation. The original-paper CodeWM protocol instead estimates pass@1 from 20 completions and detects the first valid completion.

## 4 Measured results

Table 1 preserves the complete primary table, including capped cells. For example, the CodeWM bias-2 LiveCodeBench result covers 104 of 400 tasks; it is not a full-split estimate. The bias-8 CodeWM cells are also capped. The paired quality baseline belongs to each row's own task cohort and generation configuration.

SeqMark has a discrete per-program score. Its accepted-score group may exceed a low false-positive budget, causing zero attainable true-positive rate at that budget even when AUROC is high. The report separately retains the training-selected classification threshold, its precision and recall, and the full empirical receiver operating characteristic. Those quantities do not replace the deterministic fixed-FPR columns.

Table 1: Qwen3.5-4B baseline measurements. HE is HumanEval; LCB is LiveCodeBench. $n$ is the evaluated task cohort, Scored is the watermarked score count, and Unsc. counts extraction failures. $\text{TPR}_1$ and $\text{TPR}_5$ use 1% and 5% false-positive budgets. P@1 is pass@1; + and − denote watermarked and paired unwatermarked programs. Rows use one generation seed.

| Method | Impl. | Data | $n$ | Scored | Unsc. | AUROC | $\text{TPR}_1$ | $\text{TPR}_5$ | $\bar{z}^+$ | $\bar{z}^-$ | P@1$^+$ | P@1$^-$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| KGW | pilot | HE | 164 | 147 | 17 | .969 | 91.8% | 93.2% | 30.92 | -0.82 | 0.6% | 76.2% |
| KGW | pilot | MBPP | 500 | 257 | 243 | 1.000 | 99.6% | 100.0% | 34.15 | -0.90 | 0.2% | 54.4% |
| KGW | pilot | LCB | 400 | 287 | 113 | .675 | 41.8% | 52.3% | 8.38 | -0.23 | 1.0% | 43.8% |
| KGW | official | HE | 164 | 164 | 0 | .912 | 73.8% | 79.3% | 3.21 | -0.15 | 9.1% | 76.2% |
| KGW | official | MBPP | 500 | 500 | 0 | .991 | 93.8% | 95.6% | 4.51 | -0.13 | 4.2% | 54.4% |
| KGW | official | LCB | 400 | 387 | 13 | .996 | 94.1% | 97.4% | 6.86 | -0.73 | 0.8% | 44.3% |
| SWEET | port | HE | 164 | 164 | 0 | .765 | 22.6% | 43.3% | 1.88 | 0.44 | 63.4% | 79.3% |
| SWEET | port | MBPP | 500 | 500 | 0 | .828 | 36.2% | 51.6% | 3.23 | 0.51 | 44.8% | 59.2% |
| SWEET | port | LCB | 400 | 373 | 27 | .870 | 63.3% | 70.8% | 11.43 | 1.36 | 27.0% | 42.5% |
| SemStamp | pilot | HE | 164 | 164 | 0 | .546 | 3.7% | 8.5% | 0.50 | 0.24 | 73.2% | 76.2% |
| SemStamp | pilot | MBPP | 500 | 499 | 0 | .560 | 2.6% | 10.8% | 0.57 | 0.13 | 57.0% | 54.4% |
| SemStamp | pilot | LCB | 400 | 386 | 14 | .709 | 28.0% | 38.1% | 3.40 | 0.65 | 43.5% | 43.8% |
| SemStamp | official | HE | 164 | 164 | 0 | .545 | 1.2% | 4.9% | 0.54 | 0.35 | 76.2% | 76.2% |
| SemStamp | official | MBPP | 500 | 500 | 0 | .512 | 2.0% | 4.8% | 0.45 | 0.41 | 56.0% | 54.4% |
| SemStamp | official | LCB | 400 | 392 | 8 | .651 | 15.6% | 24.7% | 1.87 | 0.76 | 43.8% | 43.8% |
| CodeWM | port, $\delta = 2$ | HE | 164 | 164 | 0 | .577 | 1.2% | 6.1% | -0.01 | -0.28 | 73.8% | 76.2% |
| CodeWM | port, $\delta = 2$ | MBPP | 500 | 500 | 0 | .620 | 4.2% | 11.2% | 0.01 | -0.39 | 55.2% | 56.8% |
| CodeWM | port, $\delta = 2$ | LCB | 104 | 96 | 8 | .897 | 42.7% | 56.2% | 3.06 | -1.09 | 41.3% | 45.2% |
| CodeWM | port, $\delta = 8$ | HE | 88 | 88 | 0 | .797 | 15.9% | 39.8% | 3.32 | -0.10 | 50.0% | 84.1% |
| CodeWM | port, $\delta = 8$ | MBPP | 96 | 96 | 0 | .889 | 55.2% | 61.5% | 4.28 | -0.35 | 20.8% | 68.8% |
| CodeWM | port, $\delta = 8$ | LCB | 48 | 45 | 3 | .967 | 75.6% | 88.9% | 12.59 | -1.07 | 10.4% | 47.9% |
| SeqMark | Gemma | HE | 164 | 164 | 0 | .933 | 0.0% | 91.5% | 2.72 | -0.17 | 40.9% | 48.8% |
| SeqMark | Gemma | MBPP | 500 | 500 | 0 | .929 | 0.0% | 0.0% | 2.83 | -0.03 | 29.6% | 35.2% |
| SeqMark | SBERT | HE | 164 | 164 | 0 | .905 | 0.0% | 0.0% | 2.63 | -0.07 | 39.6% | 48.8% |
| SeqMark | SBERT | MBPP | 500 | 500 | 0 | .909 | 0.0% | 0.0% | 2.73 | -0.00 | 29.2% | 35.2% |

## 5 Original-paper CodeWM reproduction

The completion-prompt reproduction uses DeepSeek-Coder-1.3B-Instruct and OpenCoder-1.5B-Instruct. The always-open switch ($-10$) and gated switch (0.7) remain separate settings. Table 2 uses paired unwatermarked detection negatives. The historical human-reference detection comparison is retained separately in the detailed repository report. The historical training corpus used the same MBPP test prompts as upstream; this convention differs from the disjoint chat-protocol training corpus.

Table 2: CodeWM completion-prompt reproduction. All rows use the full named benchmark split. Quality uses the 20-sample estimator, and detection uses the first valid completion. TPR columns and branch signs follow Table 1.

| Model | Switch | Data | $n$ | AUROC | $\mathrm{TPR}_1$ | $\mathrm{TPR}_5$ | $\bar{z}^+$ | $\bar{z}^-$ | $\mathrm{P@1}^+$ | $\mathrm{P@1}^-$ |
|---|---|---|---|---|---|---|---|---|---|---|
| DeepSeek | −10 | HE | 164 | .690 | 3.0% | 20.1% | 0.52 | -0.41 | 53.0% | 58.0% |
| DeepSeek | −10 | MBPP | 500 | .708 | 7.0% | 22.0% | 0.51 | -0.48 | 34.5% | 39.4% |
| OpenCoder | −10 | HE | 164 | .655 | 3.0% | 16.5% | 1.31 | 0.32 | 59.6% | 67.0% |
| OpenCoder | −10 | MBPP | 500 | .732 | 9.4% | 25.8% | 1.12 | 0.17 | 39.8% | 42.6% |
| DeepSeek | 0.7 | HE | 164 | .727 | 4.9% | 19.5% | 0.70 | -0.11 | 56.0% | 58.3% |
| DeepSeek | 0.7 | MBPP | 500 | .683 | 8.4% | 21.0% | 0.78 | 0.10 | 37.7% | 39.4% |
| OpenCoder | 0.7 | HE | 164 | .691 | 0.6% | 7.9% | 0.92 | 0.36 | 64.7% | 66.8% |
| OpenCoder | 0.7 | MBPP | 500 | .689 | 2.8% | 7.0% | 0.96 | 0.33 | 42.6% | 43.5% |

# 6 Limitations

The measurements retain their original generation settings and task cohorts. Capped CodeWM cells and cancelled SeqMark LiveCodeBench executions do not supply missing full-split measurements. Scoreable-subset detection cannot be interpreted without the extraction and null-score counts. Different prompting, quality estimators, sampling, and model families prevent pooling the original-paper reproduction, the first round, and the main comparison. No new benchmark execution or method-selection criterion is introduced by this report.

# A Evidence and reproducibility

The complete Markdown report is `out/README.md`. It preserves parameter searches, threshold classification, figures, historical comparisons, and superseded values. `results/experiments.csv` indexes generation artifacts; `results/inventory.json` maps original paths to archives and file hashes. The archives preserve original bytes, including per-task verdicts and recorded SeqMark manifests. Missing historical provenance remains unavailable.

```
python3 src/result_archive.py verify
python3 src/result_archive.py restore
cd src
python3 utils.py summarize-methods
python3 utils.py summarize-seqmark
```

The original SemStamp LiveCodeBench control quality cells reported 44.3%. The saved control texts match all 400 KGW pilot controls, whose evaluator records 175 passes (43.75%). Table 1 therefore reports 43.8%; the correction record retains the original display and source hashes.

Historical baseline comparisons and conventions are retained in the detailed Markdown report and the evidence archives. The CodeWM implementation and completion-prompt reproduction protocol remain in the source tree. All documents are maintained locally.